Text-IRSTD: Leveraging Fuzzy Text to Promote Infrared Small Target Detection in Complex Scenes

Liqiong Chen, Yuhao Liu, Zhaobing Qiu, Shuyuan Zheng, Jiaxin Zou, Yangping Yuan, and Feng Huang

***Abstract*—Infrared small target detection (IRSTD) has become a hot research topic in computer vision due to its wide applications and technical challenges. Existing methods usually focus on mining limited visual features of targets, which struggle to cope with complex and diverse detection scenarios. Although some researchers have attempted to introduce textual information for improved detection, explicit text prompts create a cross-modal semantic gap, resulting in suboptimal detection performance. To address the above issues, we introduce a novel approach leveraging fuzzy text to guide IRSTD, called Text-IRSTD. It expands classical IRSTD to fuzzy text-guided IRSTD, providing a new research idea. In particular, we first devise a novel fuzzy semantic text prompt that adapts to ambiguous target categories in images and bridges the semantic gap between texts and images. Then, we devise a text-guided CNN-Mamba hybrid encoder (TCMHE), which can adaptively capture key features from both global and local domains. Finally, a progressive cross-modal semantic interaction decoder (PCSID) is proposed to facilitate deeper information fusion between texts and images. Furthermore, we construct a new benchmark dataset, FZDT, comprising 7230 multiscenario infrared (IR) images accompanied by fuzzy semantic text annotations. Extensive experimental results demonstrate that Text-IRSTD outperforms the state-of-the-art, particularly in unseen detection scenarios.**

***Index Terms*—Convolutional neural network (CNN)-mamba, cross-modal semantic interaction, fuzzy text, infrared small target detection (IRSTD).**

I. Introduction

As a hotspot and challenging issue in computer vision, infrared small target detection (IRSTD) plays an increasingly vital role across multiple domains such as low-altitude economy and military early warning [1], [2], [3], [4]. However, compared to visible light images, infrared (IR) images typically lack texture and detailed information. Furthermore, due to the long-range detection needs, imaged targets in IRSTD typically occupy only a small number of pixels, further complicating target detection [5], [6], [7]. Therefore, how to accurately detect small IR targets from complex and diverse scenes remains an open issue.

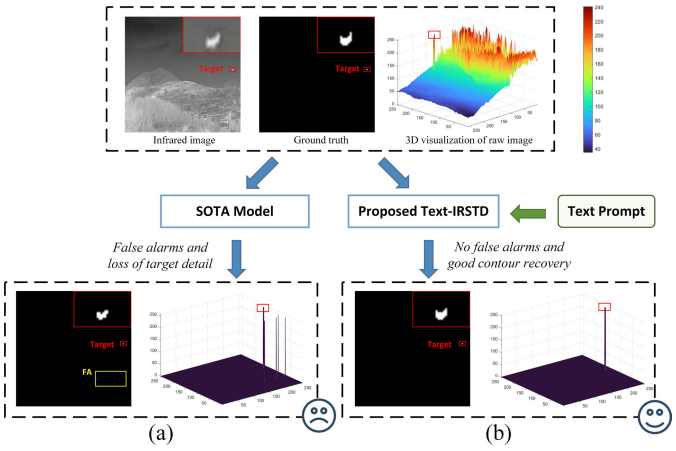


Fig. 1. Visual comparison between the SOTA model and proposed Text-IRSTD. (a) Detection results of the SOTA model [8], which relies only on visual features. (b) Detection results of the proposed Text-IRSTD, which uses both textual and visual features.

In the early days, many traditional methods emerged to improve detection accuracy. They mainly consist of filter-based methods [9], [10], local feature representation-based methods [11], [12], [13] and sparse representation-based methods [14], [15], [16]. However, reliance on artificial feature representations prevents them from further mining the deep semantic information of real targets. On the contrary, deep learning (DL)-based methods autonomously learn multilayered features in a data-driven manner and achieve significant performance improvements. For example, Dai et al. [17] propose an asymmetric contextual modulation (ACM) module to better interact with high-level semantics and low-level detail features. For higher feature utilization, Zhang et al. [18] propose a hierarchical attention fusion network (HAFNet), which mines the correlations between multilevel feature maps. In addition, researchers have also attempted to fuse prior information [19], [20] or capture long-range dependencies [21], [22] for higher detection accuracy.

However, limited by scarce image features, existing methods fail to perform satisfactorily in complex scenes. As shown in Fig. 1, the SOTA model still suffers from many false detections due to limited ability to discriminate between targets and interference. Recently, with the development of large-scale visual language models [23], semantic text–image-

Received 1 July 2026; revised 9 August 2026; accepted 19 August 2026. Date of publication 26 August 2026; date of current version 16 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62405060, in part by the Natural Science Foundation of Fujian Province under Grant 2022J05113 and Grant 2024J01245, and in part by the Educational Research Program for Young and Middle-Aged Teachers of Fujian Province under Grant JAT210035. *(Corresponding authors: Zhaobing Qiu; Yangping Yuan; Feng Huang.)*

Liqiong Chen, Yuhao Liu, Zhaobing Qiu, Shuyuan Zheng, Jiaxin Zou, and Feng Huang are with the School of Mechanical Engineering and Automation, Fuzhou University, Fuzhou 350108, China (e-mail: qiuzhaobing@fzu.edu.cn; huangf@fzu.edu.cn).

Yangping Yuan is with Zhipu AI, Beijing 100190, China (e-mail: yangping001.yuan@aminer.cn).

Digital Object Identifier 10.1109/TGRS.2026.3727675

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

based models [24], [25] have made impressive progress in generic target detection. However, applying them directly to IRSTD does not work well. For example, Zhang et al. [26] first attempted to design a detailed text prompt incorporating object category, shape, and position to establish text–image cross-modal correlations for IRSTD. However, IR small targets typically occupy only a few pixels with ambiguous categories, and detailed information such as location cannot be obtained prior to object detection, which limits their performance and applications. Subsequently, Zhang et al. [27] propose using generative pretrained transformer (GPT) to generate text prompts that describe only the background, thereby achieving better text–image matching by ignoring target information. However, on the one hand, this approach overlooks important target clues, leading to performance bottlenecks; on the other hand, its reliance on large models to generate text descriptions limits practical detection efficiency and application scenarios.

To address the above challenges, this article proposes a novel approach leveraging fuzzy text to guide IRSTD, called Text-IRSTD. It expands classical IRSTD to fuzzy text-guided IRSTD, providing a new idea for IRSTD. As shown in Fig. 1, by utilizing both text and image information, the proposed Text-IRSTD can further suppress clutter interference and improve detection performance. On the one hand, we devise a novel fuzzy semantic text prompt to accommodate ambiguous target categories. This fuzzy prompt with "`[Interested Region] target`" preserves target cues and provides more proper text–image semantic associations, thereby further improving detection performance. Furthermore, proposed text prompts can be preconfigured based on task-specific prior knowledge, offering good adaptability and practicality. On the other hand, we propose a text-guided CNN-Mamba hybrid encoder (TCMHE) and progressive cross-modal semantic interaction decoder (PCSID) to better fuse text and image information, thereby further improving detection performance. Through text-guided global–local feature fusion (TGFF) blocks, TCMHE can adaptively capture both spatial details and long-range dependencies of targets. PCSID consists of two core components: the text-guided feature aggregation (TGFA) block and text-guided semantic interaction (TGSI) block. In particular, TGFA first utilizes text information to guide cross-layer visual feature aggregation for key feature extraction. Where key features (e.g., contrast [11]) refer to those features that are more important than others (e.g., intensity) for IRSTD. Then, the TGSI block simultaneously performs feature modulation from two different perspectives, thus realizing a more refined text–image information interaction. In addition, we construct a new benchmark consisting of 7230 IR images of different scenarios with fuzzy semantic textual annotations, called FZDT. Extensive experiments on multiple public datasets demonstrate that our approach achieves better detection performance and generalization than representative state-of-the-art algorithms.

In summary, our main contributions are as follows.

1) We propose a novel approach called Text-IRSTD, which introduces fuzzy semantic text prompts to overcome the performance bottleneck of existing IRSTD methods that rely solely on visual information. To the best of our knowledge, this is among the earliest attempts to incorporate fuzzy text prompts into IRSTD, offering a new research perspective for numerous related fields.
2) A novel fuzzy semantic text prompt is devised to accommodate ambiguous IR small target categories, providing more proper text–image associations that facilitate clutter suppression and target contour recovery.
3) We propose a text-guided convolutional neural network (CNN)-Mamba dual-branch encoder, TCMHE, which utilizes semantic text guidance to extract key features of targets from both global and local domains.
4) A progressive text–image semantic interaction decoder called PCSID is proposed, which utilizes fuzzy text prompts to guide contextual feature aggregation and multidimensional refinement of feature maps.
5) We construct a new benchmark named FZDT, comprising 7230 IR images from diverse scenes with fuzzy semantic text annotations. Extensive experiments reveal that the proposed Text-IRSTD outperforms state-of-the-art methods in both detection performance and generalization to unseen scenarios.

A preliminary version of this article appears in [28]. This article is a comprehensive extension of the conference version. We specifically restructure a novel image encoder, TCMHE, for IR small targets, rather than adopting the previously generic encoder [29]—which lacks robustness in complex scenes. The primary improvements are as follows. First, we construct a dual-branch encoder architecture based on CNN and Mamba to better capture pure local and global features of targets. Second, we propose a text-guided feature fusion block TGFF, which adaptively aggregates key features from both global and local domains. Subsequently, we apply the new approach to more extensive and challenging IRSTD tasks, verifying its advancement and generalizability. Finally, further analysis of Text-IRSTD not only confirms the stability of our method but also reveals the effectiveness of specific structures.

II. Related Work

In this section, we first provide a brief overview of existing vision-only IRSTD methods. Then, we introduce existing text–image models and related work in IRSTD.

A. Vision-Only IRSTD Methods

In the early days, many traditional methods emerged to improve detection accuracy. Filter-based methods [9], [10], [30] design specific filter structures to suppress background noise and enhance targets. Local feature representation methods [11], [12], [13] achieve target detection by constructing local feature extraction models that measure contrast differences. Sparse representation-based methods [14], [15], [16] characterize small targets as sparse components of IR images, achieving target extraction through matrix decomposition and reconstruction. These methods have achieved impressive performance improvements in specific detection scenarios. However, reliance on artificial feature representations prevents them from further mining the deep semantic information of real targets.

With rapid advances in DL technology, DL-based methods have dominated the IRSTD field by autonomously learning

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

multilevel features in a data-driven manner. To fully mine the limited visual features of targets, some researchers focus on constructing more advanced feature extraction modules. For example, Kou et al. [31] integrate multitype convolution operations to improve feature extraction performance. For better capture of multiscale information, Sun et al. [32] design multireceptive-field convolution structures to enhance the local detail features. To fully utilize frequency-domain information, Xu et al. [33] propose a hybrid-domain network (HDNet) that integrates frequency-domain features for target enhancement. Subsequently, Hu et al. [34] design a dual-transformer architecture to selectively enhance important frequency features, while Zhuang et al. [35] construct a frequency-aware attention network for robust target representation. Other researchers also focus on improving the cross-layer feature fusion to mitigate feature loss of IR small targets. For example, Dai et al. [17] propose an ACM module to better interact with high-level semantics and low-level detail features. In order to interact cross-layer features incrementally, Li et al. [36] propose a dense nested attention network (DNA-Net). For higher feature utilization, Zhang et al. [18] propose HAFNet, which mines the correlations between multilevel feature maps. To capture complementary features across different layers, Yue et al. [37] design a shallow–deep synergistic detection network (SDS-Net).

Several researchers attempt to improve the model structure [38]. For example, Wu et al. [39] embed a smaller U-Net into the U-Net backbone to learn target features from multiple levels and scales; Fan et al. [40] introduce a diffusion-based continuous feature representation branch to enhance target details; and Zhang et al. [41] design a global–local fuzzy computation module for uncertainty-aware feature fusion. For more refined feature extraction, Guo et al. [42] propose a dual-encoder multistage fusion architecture. Furthermore, some researchers improve performance by incorporating prior information [19]. For example, Wu et al. [20] design learnable local kernel modules to enhance target salient features. In addition, some researchers also attempt to utilize the prior knowledge from the foundation model segment anything model (SAM) to improve detection performance. For example, Zhang et al. [42] capture key features of targets by embedding Perona–Malik diffusion (PMD)-based blocks, while Fu et al. [43] design a unified self-prompt learning (SPL) framework to enhance contextual understanding of IRSTD.

Recently, the Mamba architecture, based on the state-space model (SSM), has been widely applied to numerous computer vision tasks [44], [45]. For example, the high-order vision Mamba UNet (H-vmunet) [46] improves medical image segmentation accuracy by balancing global and local features, while the multiscale convolution and Mamba fusion model (MF-Mamba) [47] designs a global–local state-space (GLSS) module with multikernel parallel convolution to improve the remote sensing image segmentation performance. In addition, Mamba-based methods have also made remarkable progress in IRSTD, thanks to their efficient linear computational complexity and robust ability to model long-range dependencies. To better balance global and local features, Chen et al. [21] construct a nested "Mamba-in-Mamba" architecture, while Li et al. [22] propose a dual-stream "CNN-Mamba" architecture.

In summary, numerous methods have made impressive progress in the field of IRSTD. However, IR small targets possess extremely limited visual features, trapping them in performance bottlenecks across complex and diverse detection scenarios. To this end, we introduce textual information into IRSTD and propose a novel framework guided by fuzzy semantic text. By utilizing text–image cross-modal information, Text-IRSTD can further suppress background interference and improve detection performance.

B. Text–Image Models and Related Work in IRSTD

With the advancement of DL techniques and the support of large datasets, text–image models have achieved success [48]. In particular, contrastive language-image pre-training (CLIP) [23] utilizes a 400 million matched pairs dataset and contrastive learning techniques to demonstrate powerful performance and wide application prospects in multiple domains and tasks. CLIP significantly improves text–image alignment and avoids the reliance on explicit labeling. With the support of CLIP models, text-driven generic target detection has also developed rapidly.

For better alignment of text–image features, RegionCLIP [49] matches local image regions with text prompts to improve object detection performance. To improve generalization across diverse scenarios, ViLD [50] utilizes knowledge distillation techniques to transfer knowledge from pretrained open-vocabulary image classification models into two-stage detectors. To facilitate interaction between visual and textual information, YOLO-World [51] designs a reparameterizable visual–language path aggregation network and region-text contrastive loss. In addition, OW-OVD [52] proposes a visual similarity-based feature selection and hybrid feature-uncertainty fusion method to construct a unified open-world and open-vocabulary object detection framework.

Although existing text–image models have made significant progress in general object detection, they do not perform well in IRSTD tasks. To establish the correlation between text and images for IRSTD, Zhang et al. [26] design a detailed text prompt that includes object category, shape, and location, which interacts with SAM via an attention mechanism. However, IR small targets typically occupy only a few pixels with ambiguous categories, and detailed information such as location cannot be obtained prior to object detection, which limits their performance and applications. To address this, Zhang et al. [27] subsequently propose using GPT to generate text prompts that describe only the background, thereby achieving better text–image matching by ignoring target information. However, on the one hand, this approach overlooks important target clues, leading to performance bottlenecks; on the other hand, its reliance on large models to generate text descriptions limits practical detection efficiency and application scenarios.

Overall, existing text–image models have made significant progress, but it remains challenging to further improve IRSTD performance. To address this, this article proposes a novel fuzzy text prompt and cross-modal interaction method that

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ACROSS IOU(%), P_d(%), $F_a(10^{-6})$, AND F_1(%). THE BEST VALUES ARE HIGHLIGHTED WITH BOLD, THE SECOND AND THIRD BEST VALUES ARE UNDERLINED. NOTE THAT "W/O TEXT" DENOTES REMOVING THE TEXT-GUIDED BRANCH, WHILE "W/O SI" DENOTES THE USE OF THE TASK DESCRIPTION

Method	NUDT-SIRST				IRSTD-1k			
	IoU	P_d	F_a	F_1	IoU	P_d	F_a	F_1
IPI [58] (2013)	38.15	90.78	369.9	61.52	30.95	78.91	180.4	55.15
PSTNN [59] (2019)	17.21	54.47	81.62	29.45	25.58	62.96	78.91	41.35
FKRW [60] (2019)	25.38	60.26	107.5	38.54	15.89	54.54	26.75	27.08
GSWLCM [61] (2022)	4.909	65.00	15.71	9.301	3.068	59.25	26.94	6.045
RUMFR [13] (2024)	26.08	67.89	96.42	39.25	10.26	53.87	25.29	18.59
ACMNet [17] (2021)	68.14	96.57	17.38	83.41	61.14	87.88	41.79	74.17
ALCNet [62] (2021)	72.56	96.05	7.572	86.78	58.27	89.90	44.69	72.92
AGPCNet [63] (2023)	85.97	97.63	7.228	93.51	62.51	91.58	20.97	77.68
DNANet [36] (2023)	94.70	98.68	3.212	97.31	64.51	90.23	18.27	78.77
UIUNet [39] (2023)	89.15	97.11	5.506	94.23	64.32	90.24	28.33	79.02
RDIAN [32] (2023)	86.08	98.15	6.711	92.36	61.86	88.56	41.44	76.27
DMFNet [42] (2024)	87.67	98.42	2.294	93.30	64.45	90.14	18.24	78.52
SeRankDet [64] (2024)	89.87	98.95	21.86	94.66	64.76	88.89	10.55	78.58
SCTrans [8] (2024)	94.33	98.68	4.532	97.08	64.66	93.27	21.29	78.53
HAFNet [18] (2025)	94.52	98.95	1.721	97.18	64.07	91.58	16.68	78.10
HDNet [33] (2025)	84.31	97.89	7.457	91.49	63.33	91.50	10.17	77.54
SDS-Net [37] (2025)	89.41	98.16	2.065	94.41	64.21	92.59	13.17	78.20
SAM-SPL [44] (2025)	94.78	98.95	4.59	97.32	68.65	90.24	20.76	81.41
DCCS-Det [22] (2026)	77.79	96.84	17.84	87.51	61.87	90.82	**8.351**	76.43
Text-IRSTD (w/o text)	94.44	98.95	3.671	97.14	63.32	91.92	45.28	77.54
Text-IRSTD (w/o SI)	95.79	98.68	3.786	97.85	65.58	91.58	27.31	79.21
Text-IRSTD	**96.57**	**99.74**	**1.434**	**98.26**	**69.74**	**93.27**	26.32	**82.17**

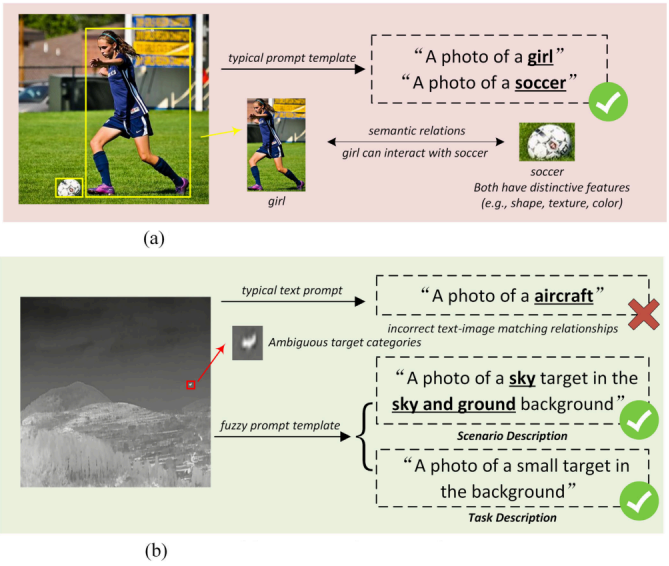


Fig. 2. (a) Typical semantic text prompt for generic target detection. (b) Proposed fuzzy semantic text prompt for IRSTD.

not only establishes more effective text–image associations to improve detection performance but also offers simplicity of implementation and good application potential.

III. FUZZY SEMANTIC TEXT PROMPT FOR IRSTD

Semantic text prompts aim to establish a proper matching relationship between text and images, which is crucial for cross-modal target detection. In generic target detection [49], [50], researchers usually directly describe specific categories of targets, such as "`A photograph of a girl`" and "`A photograph of a soccer`" in Fig. 2(a). In this way, it is possible to focus not only on the visual features of a single target but also to further explore semantic relations between multiple targets, e.g., a small round target near a girl is more likely to be a soccer. However, IR small targets typically occupy only a few pixels and lack visual features such as texture. As shown in Fig. 2(b), real targets are usually presented as bright spots that cannot be further identified as a specific category such as an aircraft or a bird. Therefore, using existing explicit text prompts to describe ambiguous IR small target categories will lead to incorrect text–image matching relationships, impairing detection accuracy and model generalizability.

To address this issue, we propose a fuzzy semantic text prompt for IRSTD to establish proper text–image associations. In this work, the fuzzy text refers to natural language prompts that intentionally avoid specifying concrete object categories (such as "aircraft" or "bird") and instead describe the target based on areas of interest and the scene context. For both effectiveness and practicality, we provide two types of fuzzy prompts: scene descriptions and task descriptions. For applications with some known information, we use the scene description and describe one IR image as "`A photograph of a [Interested Region] target in the [Scene] background`," where `[Interested Region]` denotes the region where targets need to be detected, e.g., sky target and ground target, and `[Scene]` denotes the detection scene, e.g., sky-ground background, ocean-sky background. For example, in air

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

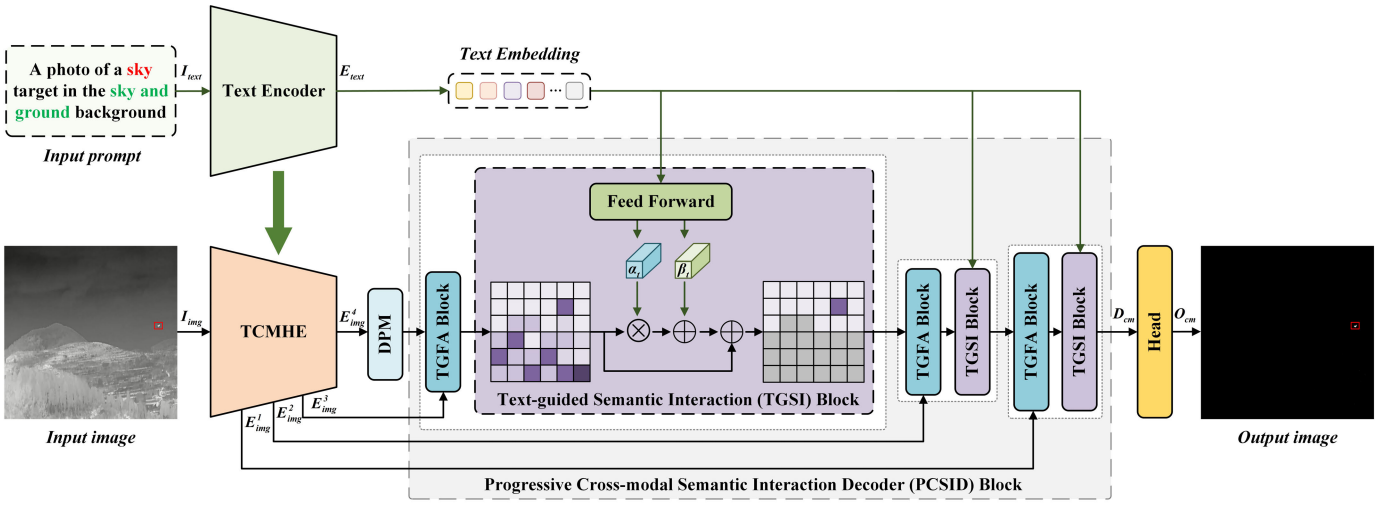


Fig. 3. Overview of the proposed text-IRSTD. It consists of three main components: text encoder, TCMHE, and cross-modal decoder PCSID, with their outputs represented as E_{text}, $E_{img}^{(i)}$, and D_{cm}, respectively. Where DPM denotes the detail perception module, TGFA Block denotes the text-guided feature aggregation block, and TGSI Block denotes the text-guided semantic interaction block.

defense applications, we describe the IR image in Fig. 2(b) as: "`A photograph of a sky target in the sky and ground background`," which not only establishes correct and robust matching relationships but also provides more semantic associations. In this way, the model can focus on more shared visual features of sky targets, such as shape contours, which helps target shape recovery and detection. Moreover, as shown in Fig. 1, more semantic associations are established and utilized to further suppress complex ground interference and improve detection accuracy. For applications without scene information (SI), we employ the task description and describe one IR image as "`A photograph of a small target in the background`." By establishing appropriate rather than erroneous text–image associations, this task description approach still achieves satisfactory detection performance (see Table V). Note that "w/o SI" in this article denotes the use of the task description.

IV. Methodology

In this section, we first present the overall pipeline of the proposed Text-IRSTD. Then, Section IV-B introduces the TCMHE encoder, which extracts important features from both global and local domains through text guidance. Finally, we introduce two core components of the PCSID decoder: the TGFA module (Section IV-C) for aggregating key visual features, and the TGSI module (Section IV-D) for facilitating text–image information interaction.

A. Pipeline Overview

Fig. 3 illustrates the overall pipeline of Text-IRSTD. First, an IR image I_{img} is fed into the TCMHE $\mathcal{F}_i$, while the corresponding text prompt I_{text} is directly fed into the frozen CLIP text encoder $\mathcal{F}_t$. The outputs of the two branches can be formulated as follows:

$$\begin{aligned} E_{img} &= \mathcal{F}_i\left(I_{img}\right) \\ E_{text} &= \mathcal{F}_t\left(I_{\text{text}}\right) \end{aligned} \tag{1}$$

where E_{img} denotes the visual features extracted by TCMHE and E_{text} denotes the text encoder output. In particular, TCMHE adopts a text-guided CNN-Mamba hybrid architecture to capture local spatial details and long-range dependencies, respectively. At each stage, the TGFF module is used to adaptively fuse the dual-branch features, thereby enhancing the representation capability of the encoder. $\mathcal{F}_t$ denotes the pretrained language model CLIP with frozen weights.

Then, E_{img} and E_{text} are fed into PCSID for text–image feature interaction and enhancement. The result of cross-modal interaction D_{cm} can be expressed as

$$D_{cm} = \text{PCSID}\left(E_{img}, E_{text}\right) \tag{2}$$

where PCSID consists of three groups of TGFA blocks and TGSI blocks in series. TGFA and TGSI will be introduced in Sections IV-C–IV-D, respectively. Note that the fourth layer output $E_{\text{img}}^{(4)}$ of $\mathcal{F}_i$ is enhanced by the detail perception module (DPM) and then fed to PCSID.

Finally, we apply a 1×1 convolution f_c^1 and sigmoid function to the decoder outputs D_{cm}. Thus, the cross-modal detection result O_{cm} can be obtained as follows:

$$O_{cm} = \text{Sigmoid}\left(f_c^1\left(D_{cm}\right)\right). \tag{3}$$

B. Text-Guided CNN-Mamba Hybrid Encoder

Feature extraction by encoders is crucial for target detection, particularly for IR small targets with only a few pixels. Although many improved encoders have been proposed recently, they typically rely on limited visual information, struggling to accurately extract key features of targets [22], [36]. To this end, we introduce text-guided feature extraction and propose the CNN-Mamba hybrid encoder TCMHE. Unlike existing CNN-Mamba hybrid models [22], TCMHE decouples the CNN and Mamba branches to avoid feature contamination caused by early fusion, while dynamically modulating global and local features using text semantics to enhance scene adaptability.

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

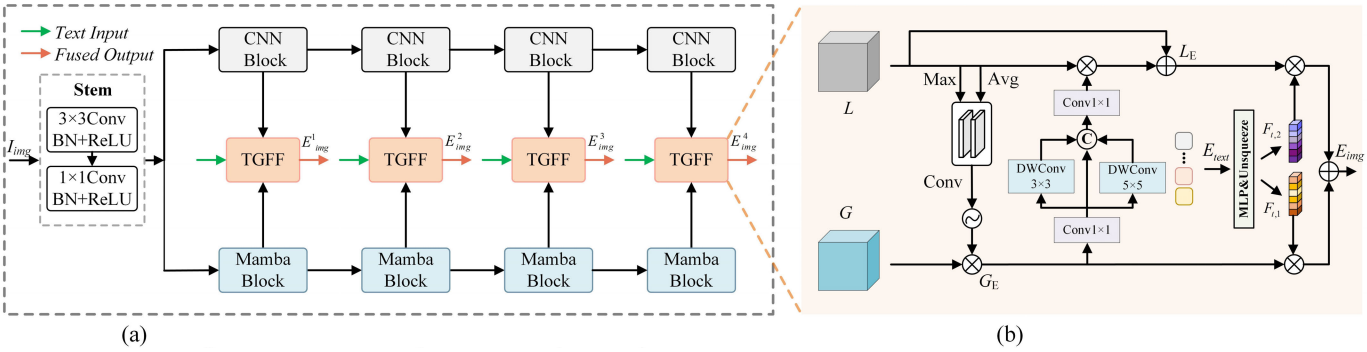


Fig. 4. Overall architecture of the proposed (a) TCMHE. Where (b) TGFF denotes the text-guided global–local feature fusion block, Conv denotes the convolution operation, BN denotes BatchNorm, ReLU denotes the ReLU activation function, and DWConv denotes depthwise convolution.

Through semantic text guidance, TCMHE can adaptively capture spatial details and long-range dependencies, further improving detection accuracy and robustness.

As shown in Fig. 4(a), TCMHE employs a dual-branch CNN-Mamba architecture to extract pure local and global features, respectively. Unlike existing two-branch architectures [22], TCMHE avoids mixing CNN and Mamba features during the feedforward process, thereby better leveraging text guidance to aggregate key features. The TGFF block then utilizes text guidance to adaptively aggregate key features from both local and global domains. In particular, an input IR image I_{img} is first fed into the stem layer to generate an initial feature representation. Then, at each encoding layer, the local feature L and global feature G of targets are extracted using a CNN block [8] and an Mamba block [21], respectively. Finally, we can obtain the fused feature output of the kth layer as follows:

$$E_{img}^{k} = \mathrm{TGFF}\left(L^{k}, G^{k}\right), k \in [1, 4] \tag{4}$$

where TGFF$(\cdot)$ denotes TGFF, and L^k and G^k denote the outputs of the CNN and Mamba blocks, respectively.

As shown in Fig. 4(b), the TGFF block is devised to effectively aggregate global and local features. In particular, TGFF first performs average pooling and max pooling on the CNN features L, and obtains the spatial attention map A through convolutional mapping

$$A = \sigma\left(\mathcal{F}_s\left(\left[\mathrm{AvgPool}\left(L\right), \mathrm{MaxPool}\left(L\right)\right]\right)\right) \tag{5}$$

where $\mathcal{F}_s(\cdot)$ denotes the convolution operation, $\sigma(\cdot)$ denotes the sigmoid activation function. Then, we obtain the enhanced Mamba feature representation G_E as follows:

$$G_{\mathrm{E}} = A \otimes G \tag{6}$$

where $\otimes$ denotes elementwise multiplication.

Subsequently, we can obtain the enhanced CNN features as follows:

$$L_{\mathrm{E}} = B \otimes L + L \tag{7}$$

where B denotes the global weighted attention map, which is defined as follows:

$$B = f_c^1\left(\mathrm{Cat}\left(f_c^1\left(G_{\mathrm{E}}\right), f_d^3\left(f_c^1\left(G_{\mathrm{E}}\right)\right), f_d^5\left(f_c^1\left(G_{\mathrm{E}}\right)\right)\right)\right) \tag{8}$$

where Cat$(\cdot)$ denotes channelwise concatenation, f_c^1 denotes 1×1 convolution, f_d^3 and f_d^5 denote 3×3 and 5×5 depthwise convolutions, respectively.

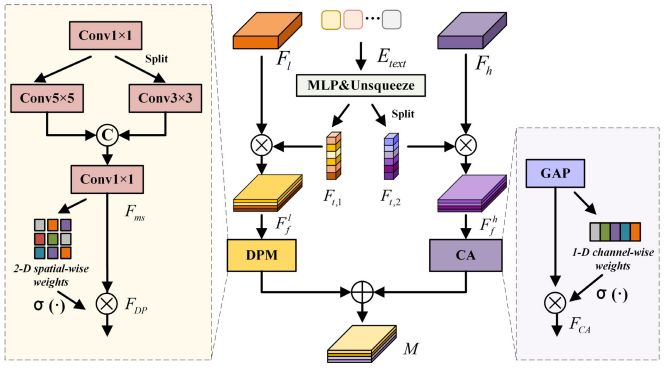


Fig. 5. Structure of TGFA block, where DPM denotes detail perception module, and CA denotes channel attention.

Finally, we utilize semantic text to adaptively aggregate the key global and local features

$$E_{img} = F_{t,1} \otimes G_{\mathrm{E}} + F_{t,2} \otimes L_{\mathrm{E}} \tag{9}$$

where $F_{t,1}$ and $F_{t,2}$ are text-guided dynamic modulation factors for global and local features, as defined below

$$F_{t,1}, F_{t,2} = \mathrm{Split}\left(\mathrm{MLP}\left(E_{text}\right)\right) \tag{10}$$

where E_{text} denotes the input text feature, multilayer perceptron (MLP) denotes the multilayer perceptron [53], and Split$(\cdot)$ denotes splitting the feature into two equal groups along the channel dimension,

C. TGFA Block

Effective aggregation of visual features is crucial for detection models. Limited by scarce visual information, existing methods struggle to capture key features from high-level and low-level feature maps. To this end, we propose a text-guided cross-layer feature aggregation block, TGFA. Unlike existing methods [36], [39], TGFA incorporates text priors to enable dynamic context interaction, thereby enhancing key features and improving model robustness. As shown in Fig. 5, TGFA transforms text features E_{text} into modulation information through an MLP as in (10), thereby guiding the model to capture key features that are easily neglected in complex scenes. The details are as follows:

$$F_f^l = F_l \otimes F_{t,1}, F_f^h = F_h \otimes F_{t,2} \tag{11}$$

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

where F_l and F_h denote low-level feature and high-level feature, F_f^l and F_f^h denote the fusion features after text modulation.

Low-level features contain rich target locations and details, but are easily lost at deep layers because small IR targets occupy only a few pixels. Therefore, we devise the DPM as shown in Fig. 5. In particular, for a given low-level fusion feature F_f^l, a 1×1 convolution is first applied to expand the channel dimension by a factor of μ, and then divide the resulting feature map into two groups $F_{l,1} \in \mathbb{R}^{\mu C/2 \times H \times W}$ and $F_{l,2} \in \mathbb{R}^{\mu C/2 \times H \times W}$ along the channel dimension.

Then, we use 3×3 and 5×5 depthwise convolutions to capture important local features such as contrast at different scales and concatenate these two groups of features along the channel dimension. The multiscale detailed feature mapping F_{ms} is obtained from the following equation:

$$F_{l,1}, F_{l,2} = \text{Split}\left(f_c^1\left(F_f^l\right)\right) \tag{12}$$

$$F_{ms} = f_c^1\left(\text{Concat}\left(f_{dwc}^3\left(F_{l,1}\right), f_{dwc}^5\left(F_{l,2}\right)\right)\right) \tag{13}$$

where Concat$(\cdot)$ denotes concatenating the two groups of features along the channel dimension, f_c^1 denotes 1×1 convolution, f_{dwc}^3 and f_{dwc}^5 denote 3×3 and 5×5 depthwise convolutions.

After obtaining the multiscale detailed feature mapping F_{ms}, we use pointwise convolution to aggregate the contextual features at each spatial location. For simplicity, it can be described as

$$F_{DP} = \sigma\left(\text{PWConv}_2\left(\gamma\left(\text{PWConv}_1\left(F_{ms}\right)\right)\right)\right) \otimes F_{ms} \tag{14}$$

where PWConv_1 and PWConv_2 denote the pointwise convolution with kernel sizes of $C/4 \times C \times 1 \times 1$ and $C \times C/4 \times 1 \times 1$, σ, and γ represent the sigmoid activation function and rectified linear unit.

Moreover, high-level features can provide more accurate semantic information while adjusting the weights of different channels can adaptively enhance the representation of key information and suppress redundant features. Therefore, for a given high-level fusion feature F_f^h, we first use global average pooling on each channel to obtain the global feature representation $z_c \in \mathbb{R}^{C \times 1 \times 1}$, which can be described as

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} F_f^h\left[:, i, j\right]. \tag{15}$$

Then, a nonlinear mapping of this global information is performed through two fully connected layers to establish dependencies between the channels.

Finally, the channel feature maps are normalized using the sigmoid function, and an elementwise multiplication is performed with the F_f^h; the output of CA is defined as

$$F_{CA} = \sigma\left(Fc_2\left(Fc_1\left(z_c\right)\right)\right) \otimes F_f^h \tag{16}$$

where $Fc_1 \in \mathbb{R}^{C/r \times C}$ and $Fc_2 \in \mathbb{R}^{C \times C/r}$ denote the two fully connected operations, r denotes the channel reduction rate.

TABLE II
COMPARISON OF MODEL PARAMETERS, FLOPS, FPS, AND INFERENCE TIME ON THE NUDT-SIRST DATASET

Model	Params (M)	FLOPs (G)	FPS (f/s)	Time (ms)
DNANet	4.70	14.26	26.59	37.61
AGPCNet	12.36	43.18	9.44	105.93
UIUNet	50.54	54.43	24.34	41.08
SCTrans	11.19	20.24	21.02	47.57
SeRankDet	108.9	140.9	73.42	13.62
HAFNet	13.55	24.68	17.70	56.49
Text-IRSTD (w/o text)	15.95	50.73	26.58	37.62
Text-IRSTD	75.41	58.49	25.14	39.77

D. TGSI Block

Due to limited visual information and the challenging text–image matching in IRSTD, existing models struggle to effectively suppress complex interference and recover target contours. Therefore, we introduce fuzzy text and propose the TGSI block, which employs two semantic parameters (gain and offset) to collaboratively suppress interference, thereby further improving detection performance. Fig. 3 shows the detailed structure of the TGSI block. First, we take the image feature M and the text feature E_{text} as inputs, where E_{text} is obtained by feeding the text prompt T_{text} into the CLIP. Notably, we freeze the weights of the text encoder to preserve the original capabilities of the pretrained model.

Furthermore, we design a feed-forward network (FFN) to further explore the deep semantic information embedded in the text features. This helps the model capture more fine-grained semantic associations between the text, the target, and the regions of interest. In particular, the text feature E_{text} is mapped into two learnable semantic parameters α_t and β_t after going through the FFN, which can be defined as

$$\alpha_t,\ \beta_t = \text{FFN}\left(E_{text}\right). \tag{17}$$

Finally, the semantic parameters α_t and β_t modulate the image feature M from the perspective of gain control and offset adjustment, respectively. In particular, the gain parameter α_t directs the network to pay more attention to the regions of interest of the text prompt and reduces false alarms in other regions. The offset parameter β_t provides finer tuning and helps in the recovery of target contour details. The fused semantic features $\hat{M}$ are obtained by

$$\hat{M} = (1 + \alpha_t) \otimes M + \beta_t. \tag{18}$$

Overall, since different channels encode different feature patterns (e.g., some channels respond strongly to sky textures, while others respond strongly to ground clutter), the fuzzy text can guide the model to enhance targets in regions of interest. In particular, the gain α_t amplifies channels consistent with the text (e.g., the sky) while suppressing other channels (e.g., the ground); the offset β_t restores target boundaries by enhancing weak signals. Consequently, the proposed TGSI achieves better target contour recovery and background suppression.

V. EXPERIMENTS

In this section, we first introduce evaluation metrics and experimental setup, including the datasets, baselines, and

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

TABLE III

DETECTION PERFORMANCE OF DIFFERENT METHODS IN UNSEEN SCENARIOS. THE FIRST SET OF EXPERIMENTS EMPLOYS NUDT-SIRST AS THE TRAINING DATASET AND NUAA-SIRST AS THE TEST DATASET; THE SECOND SET OF EXPERIMENTS EMPLOYS IRSTD-1K AS THE TRAINING DATASET AND SIRST-UAVB AS THE TEST DATASET; WHILE THE THIRD SET OF EXPERIMENTS EMPLOYS IRSTD-1K AS THE TRAINING DATASET AND IRSTD-AIR AS THE TEST DATASET

Method	NUDT-SIRST (Train) NUAA-SIRST (Test)				IRSTD-1k (Train) SIRST-UAVB (Test)				IRSTD-1k (Train) IRSTD-Air (Test)			
	IoU(%)	$P_d(\%)$	$F_a(10^{-6})$	$F_1(\%)$	IoU(%)	$P_d(\%)$	$F_a(10^{-6})$	$F_1(\%)$	IoU(%)	$P_d(\%)$	$F_a(10^{-6})$	$F_1(\%)$
ACMNet [17]	50.99	84.40	55.46	67.54	4.67	30.48	196.79	8.92	7.78	13.46	172.27	14.43
ALCNet [62]	50.94	90.83	109.05	67.50	6.99	23.93	83.74	13.09	10.85	30.21	166.06	19.57
DNANet [36]	21.36	64.22	454.79	35.20	8.89	33.16	79.22	16.33	11.69	39.74	103.68	20.93
UIUNet [39]	51.64	86.24	34.64	68.11	7.17	29.01	100.69	13.39	10.88	24.79	77.02	19.62
RDIAN [32]	44.42	82.57	118.09	61.52	4.52	32.75	215.70	8.64	12.17	32.68	117.91	21.69
SCTrans [8]	54.26	86.24	44.20	70.35	8.28	32.22	112.22	15.29	**13.11**	43.84	51.36	**23.18**
HAFNet [18]	55.98	**91.74**	50.34	71.78	7.34	22.06	76.08	13.67	11.67	41.87	95.79	20.91
SDS-Net [37]	55.77	84.40	**19.63**	71.60	7.38	32.75	109.57	13.74	9.33	34.48	**35.20**	17.07
SAM-SPL [44]	55.70	86.24	26.61	71.55	7.05	23.23	**72.03**	13.17	2.50	17.57	151.97	4.88
Text-IRSTD	**56.61**	88.07	31.74	**72.30**	**10.03**	**33.16**	72.85	**18.23**	12.49	**47.62**	147.44	22.20

implementation details. Then, quantitative and qualitative comparative experiments between Text-IRSTD and baselines are conducted in Sections V-C and V-D. Subsequently, we further evaluate the robustness and generalization of Text-IRSTD to unseen scenarios in Section V-E. Finally, comprehensive ablation experiments and hyperparameter analysis are conducted in Sections V-F and V-G, respectively.

A. Evaluation Metrics

To comprehensively compare the detection performance of different methods, we employ several widely recognized evaluation metrics as follows.

1) *Intersection Over Union (IoU):* IoU denotes the ratio of the intersection area to the union area between predicted labels and ground truth labels. It is defined as follows:

$$\text{IoU} = \frac{\sum_{i=1}^{N} \text{TP}[i]}{\sum_{i=1}^{N} (\text{T}[i] + \text{P}[i] - \text{TP}[i])} \tag{19}$$

where N is the number of samples, TP[•] denotes the number of true positive pixels, T[•] denotes the number of ground-truth positive pixels, and P[•] denotes the number of predicted positive pixels.

2) *Probability of Detection (P_d):* P_d denotes the ratio of correctly predicted targets N_{pred} over all targets N_{all}

$$P_d = \frac{N_{\text{pred}}}{N_{\text{all}}}. \tag{20}$$

Following [36], a correct prediction indicates that the center deviation between the predicted target and ground truth is less than 3 pixels.

3) *False Alarm Rate (F_a):* F_a is the ratio of the number of falsely predicted target pixels P_{false} to the total number of pixels P_{all} in the image

$$F_a = \frac{P_{\text{false}}}{P_{\text{all}}}. \tag{21}$$

4) *$F1$-Score (F_1):* F_1 is a pixel-level evaluation metric used to comprehensively evaluate precision (Prec) and recall (Rec). It is defined as follows:

$$F_1 = \frac{2 \times \text{Prec} \times \text{Rec}}{\text{Prec} + \text{Rec}}. \tag{22}$$

In addition, we employ receiver operating characteristic (ROC) curves to comprehensively evaluate the performance of models across different thresholds. The larger area under the ROC curve indicates better performance. Therefore, the area under the curve (AUC) is also employed as a quantitative metric to assess overall detection performance.

TABLE IV

ABLATION STUDY TO EVALUATE THE TCMHE, TGFA, AND TGSI BLOCKS ON THE IRSTD-1K DATASET

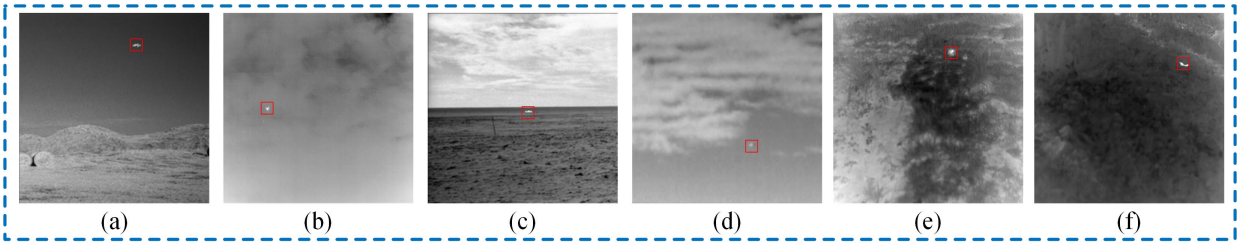
Baseline	TCMHE	TGFA	TGSI	IoU(%)	$P_d(\%)$	$F_a(10^{-6})$
✓				61.31	89.90	51.26
✓	✓			65.17	91.25	53.05
✓		✓		67.26	92.26	58.04
✓			✓	66.67	92.26	38.66
✓	✓	✓		66.50	91.25	49.38
✓	✓		✓	67.29	92.93	28.64
✓		✓	✓	68.06	90.57	38.70
✓	✓	✓	✓	**69.74**	**93.27**	**26.32**

B. Experimental Settings

1) Datasets: All experiments are conducted on the proposed FZDT dataset, which comprises 7230 multiscenario IR images and their corresponding text annotations. The image data are sourced from multiple public datasets including NUAA-SIRST [17], NUDT-SIRST [36], IRSTD-1k [6], SIRST-UAVB [54], and IRSTD-Air [55]. The corresponding text prompts are annotated according to Section III, which includes two types: scene descriptions and task descriptions. Fig. 6 provides some image–text paired data from FZDT. Taking Fig. 6(a) as an example, when detecting flying targets in a sky-ground background, the scene description is "`A photograph of a sky target in the sky and ground background`," while the task description is "`A photograph of a small target in the background`." For fair comparison, we divided each dataset of FZDT into training, validation, and test sets at a ratio of 6:2:2, following existing work [6], [56].

2) Baselines: To comprehensively evaluate detection performance, we compare proposed Text-IRSTD with state-of-the-art IRSTD methods, including five proven traditional methods (IPI [57], partial sum of the tensor nuclear norm (PSTNN) [58], facet kernel and random walker (FKRW) [59], global sparsity-weighted local contrast measure (GSWLCM) [60], and robust unsupervised multi-feature representation (RUMFR) [13]), along with 14 recent DL-based methods

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.



	Conventional prompts
(a)	A photo of a aircraft
(b)	A photo of a aircraft
(c)	A photo of a ship
(d)	A photo of a bird
(e)	A photo of a vehicle
(f)	A photo of a vehicle

	Fuzzy prompts (Scene Descriptions)
(a)	A photo of a sky target in the sky and ground background
(b)	A photo of a sky target in the sky background
(c)	A photo of a sea target in the sea and sky background
(d)	A photo of a sky target in the sky background
(e)	A photo of a ground target in the ground background
(f)	A photo of a ground target in the ground background
	Fuzzy prompts (Task Descriptions)
All	A photo of a small target in the background

Fig. 6. Examples of the proposed FZDT dataset. For each IR image, we provide the pixel-level target mask, a conventional text prompt, and the fuzzy text prompt containing both scene and task descriptions.

(ACMNet [17], ALCNet [61], AGPCNet [62], DNANet [36], UIUNet [39], receptive-field and direction induced attention network (RDIAN) [32], DMFNet [63], SeRankDet [64], SCTrans [8], HAFNet [18], HDNet [33], SDS-Net [37], SAM-SPL [43], and DCCS-Det [22]. For fair comparison, all parameters and experimental settings of baseline methods are adopted as recommended in their published papers.

3) Implementation Details: The proposed Text-IRSTD is implemented with PyTorch on an NVIDIA GeForce RTX 3090 graphics processing unit (GPU). We employ the widely used CLIP-ViT-B/32 [23] as the text encoder. The image encoder component of our Text-IRSTD does not utilize any pretrained weights. During training, each image undergoes normalization and is randomly cropped into 256×256 pixel patches. We set the batch size to 4 and the number of training epochs to 500. During training, the image and text branches simultaneously learn parameters to facilitate better cross-modal information interaction. To mitigate overfitting, we employ the AdamW optimizer with the SoftIoU loss [65]. The initial learning rate is set to 0.001 and gradually decayed to 10^{-6} using the CosineAnnealingLR scheduler, with a weight decay factor of 0.01.

C. Quantitative Results

Table I presents a quantitative comparison of different methods. Overall, DL-based methods significantly outperform traditional methods in background suppression and target contour restoration. The main reason is that traditional methods rely on manual experience modeling, which fails to deeply mine the high-level semantic information of targets. The proposed Text-IRSTD demonstrates better detection performance than all other methods. In particular, on the NUDT-SIRST dataset, Text-IRSTD outperforms the second-best SAM-SPL by 1.79%, 0.79%, and 0.94% in terms of IoU, P_d, and F_1, respectively. Surprisingly, even without any scene prior knowledge, our Text-IRSTD (w/o SI) still outperforms other methods in terms of IoU and F_1. The experiment validates the effectiveness and practicality of our novel framework based on fuzzy text prompts. On the IRSTD-1k dataset, although our method performs 17.97×10^{-6} times higher than DCCS-Det on F_a, it outperforms DCCS-Det by 7.87% and 2.45% on IoU and P_d, respectively, achieving the best detection performance. These results demonstrate that by incorporating fuzzy text prompts, our proposed Text-IRSTD effectively enhances detection performance in complex IRSTD scenarios, offering a novel approach for future research improvements.

Fig. 7 presents the ROC curves of different methods to further compare detection performance at various thresholds. It can be seen that the ROC curve of Text-IRSTD lies to the upper left of other methods, demonstrating its ability to achieve higher P_d at lower F_a. Furthermore, the AUC value of each ROC curve is also calculated for quantitative evaluation. One can see that our method achieves the maximum AUC value, implying superior overall performance. Experimental results demonstrate that by leveraging fuzzy text guidance, proposed Text-IRSTD effectively improves detection performance in complex scenarios.

Table II illustrates the comparison of the model efficiency of different methods. It can be seen that our method outperforms SeRankDet in terms of computational complexity, with relatively acceptable numbers of parameters (75.41 M) and FLOPs (58.49 G). Furthermore, Text-IRSTD achieves an inference speed of 25.14 frames per second (FPS), outperforming several methods such as SCTrans and HAFNet. When the text branch is removed, Text-IRSTD (w/o text) has only 15.95 M parameters, which is comparable to AGPCNet (12.36 M) and HAFNet (13.55 M). Thus, knowledge distillation from CLIP (e.g., TinyCLIP [66]) or replacing it with a smaller variant will significantly reduce model complexity. Furthermore, applying lightweighting techniques such as quantization and pruning will further reduce computational costs, enabling broader application in scenarios with extremely limited resources or high real-time requirements.

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

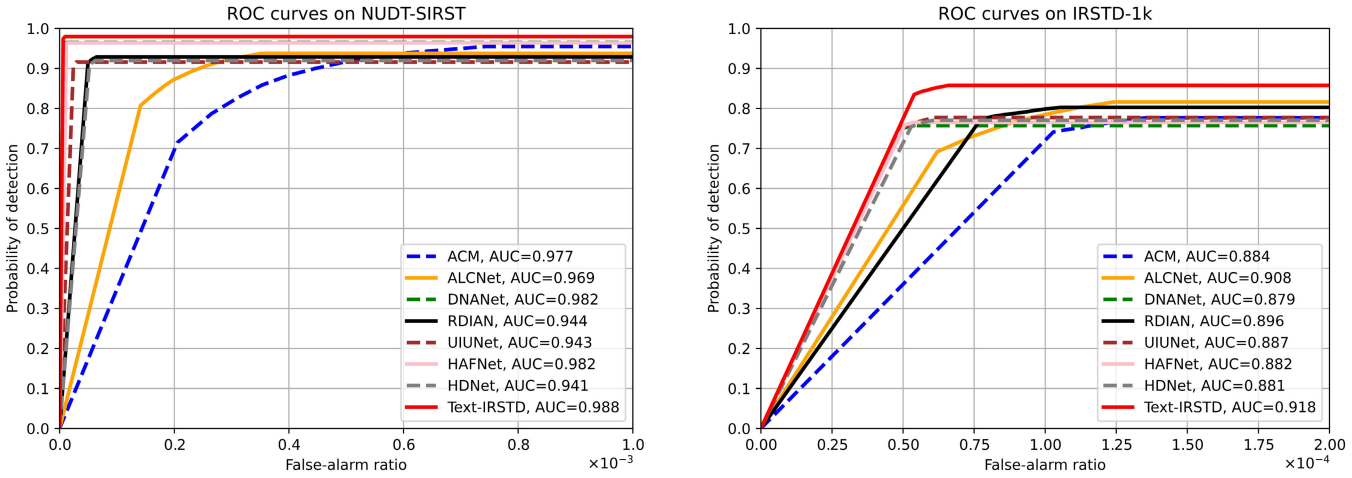


Fig. 7. ROC curves of different IRSTD methods on the NUDT-SIRST and IRSTD-1k.

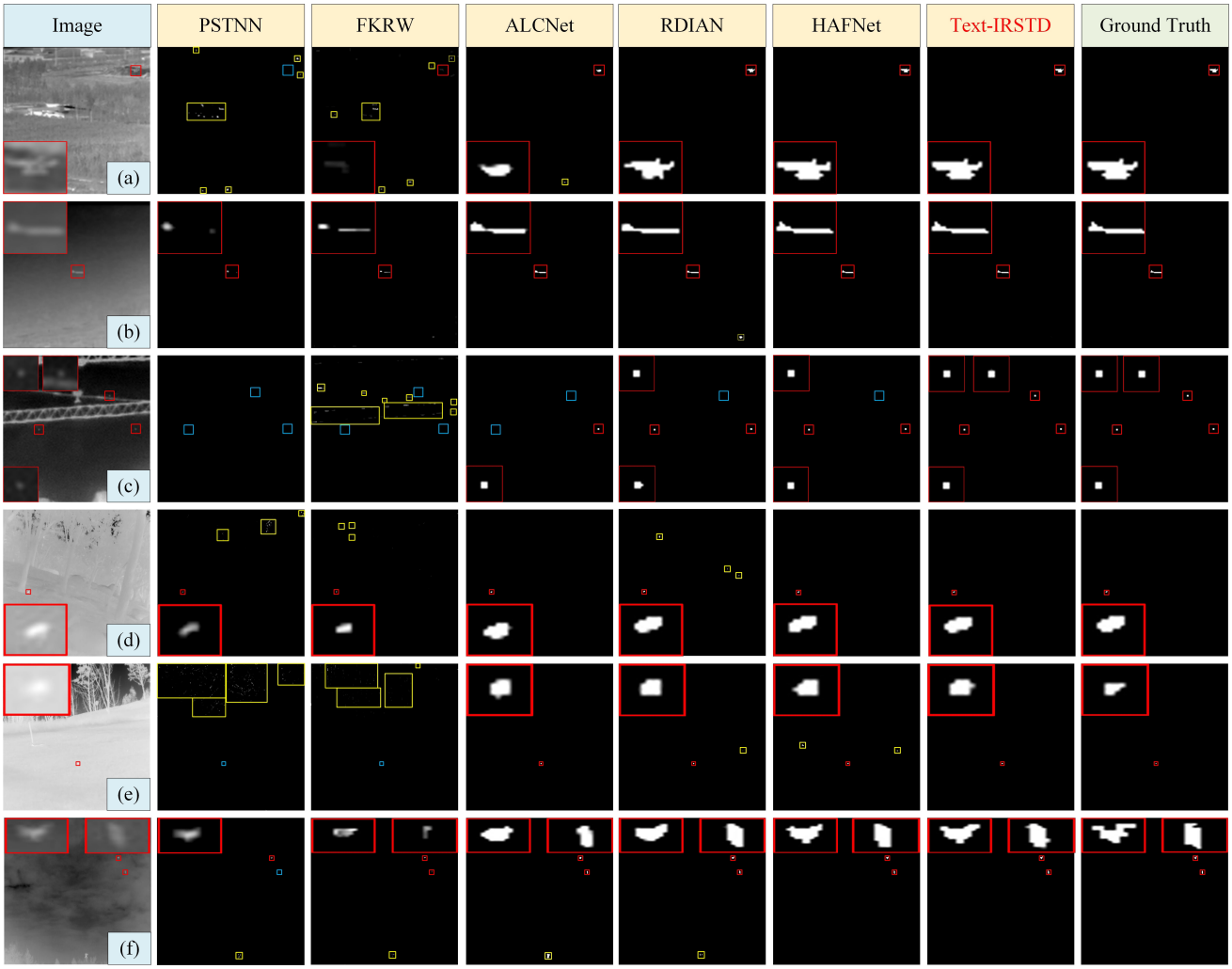


Fig. 8. Detection results of different methods on the NUDT-SIRST and IRSTD-1k datasets. The image corners present enlarged views of targets. The red, blue, and yellow rectangles denote correctly detected targets, missed targets, and false alarm targets, respectively.

D. Qualitative Results

Figs. 8 and 9 visually demonstrate the detection results of different methods on NUDT-SIRST and IRSTD-1k. Overall, our Text-IRSTD exhibits superior background suppression and target contour recovery. In particular, traditional methods (e.g., PSTNN and FKRW) struggle to effectively suppress background interference, resulting in numerous missed and false

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

TABLE V
ABLATION STUDY OF DIFFERENT INPUT PROMPTS ON THE IRSTD-1K DATASET

Type	Prompt	IoU(%)	$P_d(\%)$	$F_a(10^{-6})$
Conventional prompts	A photo of a [class]	61.73	88.55	28.01
Variants of convention	A photo of a [class] **in the [Scene] background**	64.26	89.23	32.45
Variants of Ours	**A [Interested Region] target** in the [Scene] background	<u>68.52</u>	92.93	**22.93**
Ours	**A photo of** a [Interested Region] target in the [Scene] background	**69.74**	**93.27**	<u>26.32</u>

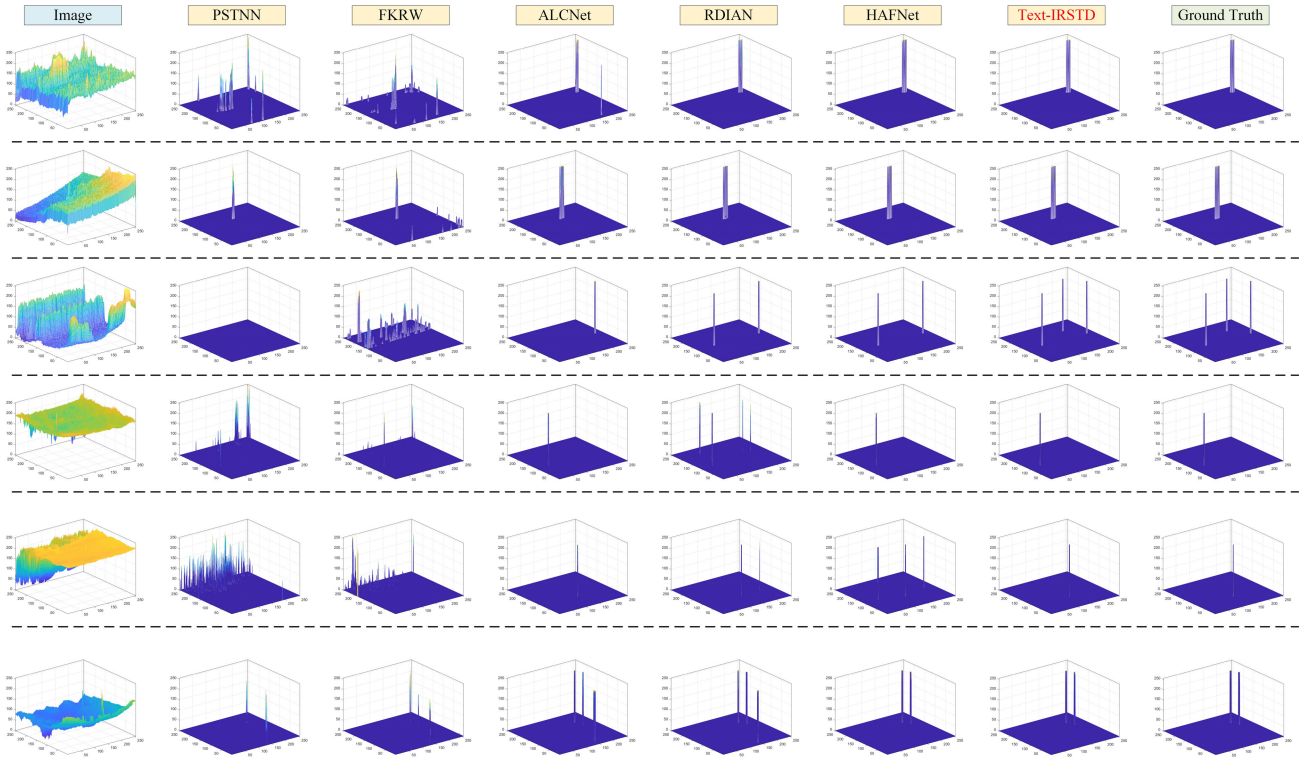


Fig. 9. 3-D visualization of salient maps of different methods on the six test images.

detections. In comparison, DL-based methods achieve better background suppression through data-driven approaches. For example, ALCNet and RDIAN suppress all background clutter in Fig. 8(a) and (b), respectively. However, compared to the recent HAFNet, they struggle to effectively preserve target contours [e.g., Fig. 8(a) and (f)]. Due to the limited visual information of IR small targets, ALCNet, RDIAN, and HAFNet miss real targets in Fig. 8(c) and exhibit a small number of false alarms in other figures. On the contrary, our method achieves higher detection accuracy and better target contour recovery. Experimental results demonstrate the effectiveness of fuzzy text guidance and text–image information interaction.

E. Performance in Unseen Scenarios

In practical applications, models often need to face unseen detection scenarios. The detection performance in unseen scenarios can evaluate the practicality and generalization of models to some extent. To further validate the advancement of Text-IRSTD, we compare the detection performance of different methods across various unseen scenarios. For experimental fairness and comprehensiveness, we conduct three sets of comparative experiments with different dataset configurations. In particular, the first set employs NUDT-SIRST as the training dataset and NUAA-SIRST as the test dataset; the second set employs IRSTD-1k as the training dataset and SIRST-UAVB as the test dataset; and the third set employs IRSTD-1k as the training dataset and IRSTD-Air as the test dataset. To better evaluate zero-shot generalization capabilities, the test set includes detection scenarios not present in the training set, as well as targets with different appearances.

Table III quantitatively compares the detection performance of different methods. Overall, the proposed Text-IRSTD achieves better comprehensive performance across all unseen scenarios. For example, our method achieves the highest IoU and F_1 scores on NUAA-SIRST while outperforming the second-best HAFNet by 0.63% in IoU. Moreover, Text-IRSTD achieves optimal values across IoU, P_d, and F_1 scores on the SIRST-UAVB dataset, demonstrating higher detection accuracy and better target contour recovery. For visual comparison, Fig. 10 presents the detection results of different methods in unseen scenes. Through improved feature fusion, recent methods SCTransNet and HAFNet achieve better target contour recovery compared to DNANet and UIUNet

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.



Fig. 10. Visual comparisons of different IRSTD methods in unseen scenarios. The enlarged view of the target is shown in image corners. The red, blue, and yellow boxes represent correctly detected targets, missed targets, and false detections, respectively.

[e.g., Fig. 10(a) and (f)]. However, they also exhibit some missed detections and numerous false alarms in Fig. 10(c)–(f). On the contrary, the proposed Text-IRSTD demonstrates better target detection performance and contour recovery in almost all images. Experimental results verify that the proposed method can effectively improve IRSTD performance in unseen scenes, offering good application potential and research value.

F. Ablation Study

TCMHE, TGFA, and TGSI are the core components of our Text-IRSTD framework. To evaluate the contributions of each component and the advancement of their internal designs, we conduct extensive ablation experiments on the IRSTD-1k dataset, which contains multiscenario IR images.

1) Performance Analysis of Each Component: To systematically evaluate the contributions of each module, we conduct ablation experiments with and without components such as TCMHE, TGFA, and TGSI. As shown in Table IV, by introducing TCMHE, our method achieves improvements of 3.86% and 1.35% in IoU and P_d, respectively. The main reason is that TCMHE can effectively capture the key global and local features, thereby improving detection performance. Furthermore, the simultaneous introduction of TGFA and TGSI improves IoU and P_d by 6.75% and 0.67%, respectively, while reducing the F_a value by 1.26×10^{-5}. This is primarily attributed to the PCSID, which enhances text–image alignment and detection performance by guiding context feature aggregation and multidimensional refinement through fuzzy text prompts. By integrating these modules, Text-IRSTD achieves significant improvements of 8.43% and 3.37% in IoU and P_d, respectively. Experimental results validate the effectiveness of each module in improving overall performance when used both individually and together.

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

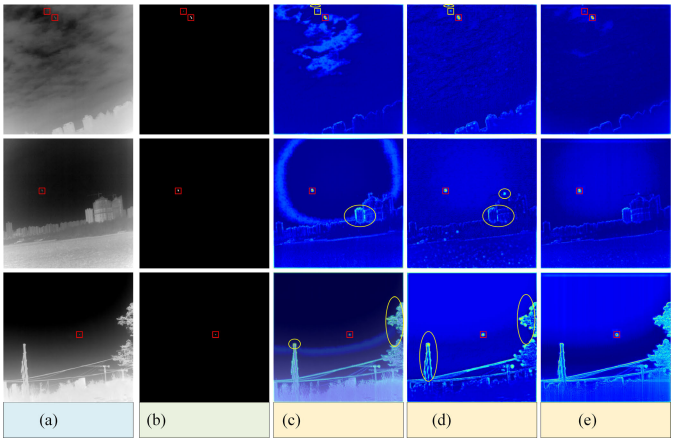


Fig. 11. Visual comparison of heatmaps for different text prompts. Where the heatmap refers to the feature map from the decoder's final layer, the red rectangles denote correctly detected targets, the yellow rectangles denote missed targets, and the yellow ellipses denote false positives. (a) Input. (b) Ground truth. (c) Conventional. (d) Variants. (e) Ours.

TABLE VI

PERFORMANCE COMPARISON OF TCMHE VARIANTS WITH DIFFERENT ARCHITECTURES. WHERE "CNN SINGLE" AND "MAMBA SINGLE" DENOTE SINGLE-BRANCH VARIANTS, WHILE "CNN DUAL" AND "MAMBA DUAL" DENOTE DUAL-BRANCH VARIANTS. "CNN-MAMBA MIX" REFERS TO THE EXISTING HYBRID ARCHITECTURE

Model	IoU(%)	P_d(%)	$F_a(10^{-6})$
CNN Single	66.16	90.24	36.65
Mamba Single	66.19	88.55	**23.12**
CNN Dual	65.80	90.91	38.56
Mamba Dual	66.32	90.24	35.03
CNN-Mamba Mix	65.85	89.56	46.38
TCMHE	**69.74**	**93.27**	26.32

2) Performance Analysis of Different Input Prompts: To validate the effectiveness of proposed fuzzy text prompts, we devise three text prompt variants for comparison. As shown in Table V, conventional text prompts with explicit target categories demonstrate the worst detection performance. The main reason is their difficulty in matching ambiguous IR small target categories. By using the `[Interested Region]` target as a text prompt, we can better establish text–image matching relationships and achieve significant improvements in IoU. Meanwhile, incorporating `[Scene]` background descriptions can further enhance text–image cross-modal semantic associations, thereby improving detection performance. In addition, adding "a photograph of" can fully leverage the text–image alignment capabilities of LLMs. Fig. 11 visually demonstrates the effectiveness of fuzzy text. When detecting sky targets in sky-ground backgrounds, conventional prompts and their variants result in more missed and false detections. On the contrary, our fuzzy text guides the model to focus on the sky region, thereby better recovering target contours and suppressing the background. Experimental results demonstrate that the proposed fuzzy text prompts effectively enhance the text–image information matching and target detection performance of IRSTD.

3) Performance Analysis of TCMHE With Different Architectures: Table VI quantitatively compares the performance of TCMHE with different architectures. It can be seen that simply increasing the number of branches does not effectively improve detection performance. Furthermore, existing CNN-Mamba hybrid architectures tend to confuse global and local features, failing to achieve satisfactory results [22]. On the contrary, the proposed TCMHE significantly improves IoU and P_d by 3.89% and 3.71%, respectively. These experimental results demonstrate the effectiveness of independent feature extraction and semantic text guidance.

TABLE VII

PERFORMANCE COMPARISON OF TGFF VARIANTS WITH DIFFERENT INTERNAL COMPONENTS. WHERE "W/O TEXT" DENOTES REMOVAL OF THE TEXT GUIDE BRANCH, "W/O L" DENOTES REMOVAL OF LOCAL FEATURE MODULATION, AND "W/O G" DENOTES REMOVAL OF THE GLOBAL FEATURE MODULATION

Model	IoU(%)	P_d(%)	$F_a(10^{-6})$
TGFF w/o text	67.14	90.91	33.67
TGFF w/o L	66.35	90.57	48.17
TGFF w/o G	67.79	91.92	29.99
TGFF	**69.74**	**93.27**	**26.32**

TABLE VIII

PERFORMANCE COMPARISON OF TGFA VARIANTS WITH DIFFERENT INTERNAL COMPONENTS. WHERE "W/O TEXT" DENOTES REMOVAL OF THE TEXT GUIDE BRANCH, "W/O CA" DENOTES REMOVAL OF THE CHANNEL ATTENTION MODULE, AND "W/O DPM" DENOTES REMOVAL OF THE DETAIL PERCEPTION MODULE

Model	IoU(%)	P_d(%)	$F_a(10^{-6})$
TGFA w/o text	65.50	92.26	33.35
TGFA w/o CA	67.07	90.24	**18.96**
TGFA w/o DPM	67.05	89.56	27.97
TGFA	**69.74**	**93.27**	26.32

4) Performance Analysis of TGFF Internal Components: To further validate the structural innovation of TGFF, Table VII presents three variants with distinct internal architectures. Without semantic text guidance, the encoder fails to adaptively aggregate global and local features, resulting in a significant 2.6% decrease in IoU. Similarly, text-guided single-branch feature modulation struggles to achieve satisfactory results. For example, when modulating only local features, TGFF w/o G still reduces IoU and P_d by 1.95% and 1.35%, respectively. Experimental results demonstrate that TGFF can better capture important spatial details and global dependencies, thereby further improving detection performance.

5) Performance Analysis of TGFA Internal Components: To further validate the structural innovation of TGFA, Table VIII presents three variants with distinct internal architectures. It can be seen that removing the w/o DPM weakens key visual features such as contrast, causing a significant 2.69% drop in IoU. In addition, removing channel attention (w/o CA) reduces the perceptual capability of visual features, causing IoU and P_d to decrease by 2.67% and 3.03%, respectively. Similarly, removing the text guide module (w/o text guide) limits cross-layer feature aggregation, causing IoU and P_d to decrease by 4.24% and 1.01%, respectively. Experimental results demonstrate that TGFA effectively guides contextual feature aggregation, thereby improving detection performance.

6) Performance Analysis of TGSI Internal Components: To validate the structural innovation of TGSI, Table IX compares four variants of text–image fusion. It can be seen that the simple concatenation operation causes significant decreases in IoU and P_d by 5.6% and 3.03%, respectively. The main

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

TABLE IX

PERFORMANCE COMPARISONS OF TGSI WITH DIFFERENT INTERNAL COMPONENTS. WHERE "CONCATENATION" DENOTES USING SIMPLE CONCATENATION OPERATIONS, "W/O TEXT" DENOTES REMOVAL OF THE TEXT GUIDE BRANCH, "W/O α_t" INDICATES REMOVAL OF GAIN CONTROL, AND "W/O β_t" DENOTES REMOVAL OF OFFSET ADJUSTMENT

Model	IoU(%)	P_d(%)	$F_a(10^{-6})$
Concatenation	64.14	90.24	46.61
TGSI w/o text	66.70	91.58	40.48
TGSI w/o α_t	67.56	90.91	45.74
TGSI w/o β_t	68.47	91.25	**24.24**
TGSI	**69.74**	**93.27**	26.32

TABLE X

PERFORMANCE COMPARISON OF DIFFERENT FREEZING STRATEGIES FOR THE CLIP TEXT ENCODER. WHERE "PARTIAL FINE-TUNING" REFERS TO FINE-TUNING THE LAST TWO LAYERS OF CLIP, AND "FULL FINE-TUNING" REFERS TO FULLY FINE-TUNING CLIP

Model	IoU(%)	P_d(%)	$F_a(10^{-6})$	F_1(%)
Partial fine-tuning	67.66	90.24	43.59	80.71
Full fine-tuning	67.15	91.25	47.54	80.35
Fully frozen	**69.74**	**93.27**	**26.32**	**82.17**

reason is that it ignores the cross-modal differences between text and images. When fine-tuning operations are introduced, detection accuracy is effectively improved. For example, introducing gain control (w/o β_t) and offset adjustment (w/o α_t) significantly increases IoU by 4.33% and 3.42%, respectively. Notably, when combining gain control and offset adjustment for fine-tuning, TGSI achieves significant improvements of 2.18% and 2.36% in IoU and P_d, respectively. This is primarily attributed to the TGSI block performing feature modulation from two different perspectives simultaneously, facilitating more refined text–image information interaction. Experimental results validate that TGSI can effectively promote cross-modal information interaction and improve detection performance.

7) Performance Analysis of CLIP Freezing Strategy: To verify the impact of different CLIP text encoder freezing strategies on text–image alignment, Table X illustrates the comparison of the experimental results of three configurations. It can be seen that both partial fine-tuning and full fine-tuning resulted in performance degradation, with IoU decreasing by 2.08% and 2.59%, respectively. The main reason is that partial fine-tuning slightly perturbed the semantic space of the CLIP pretraining, while full fine-tuning on limited IR data led to overfitting and catastrophic forgetting of general region-scene priors. On the contrary, fully freezing CLIP preserves robust prior knowledge, providing reliable region-scene semantic guidance for Text-IRSTD, thereby achieving optimal performance.

G. Core Hyperparameter Analysis

The core parameters of our method include μ and r in the TGFA block. As shown in Table XI, a smaller μ (e.g., 1.33) limits DPM's ability to extract multiscale features, while a larger μ (e.g., 3.99) introduces redundant parameters, leading to slight overfitting. Therefore, we set μ = 2.66 to achieve the optimal balance between representation capacity and generalization ability. Similarly, a smaller r (e.g., 1) results in insufficient compression, making it difficult to effectively learn interchannel dependencies, while a larger r (e.g., 8) results in excessive compression, making it easy to lose discriminative channel information. Therefore, we set r = 4 to achieve better feature representation.

TABLE XI

HYPERPARAMETER STUDY OF THE CHANNEL EXPANSION FACTOR μ AND THE CHANNEL REDUCTION RATE r ON THE IRSTD-1K DATASET

Hyper-param	IoU(%)	P_d(%)	$F_a(10^{-6})$	F_1(%)
The channel expansion factor of DPM				
$\mu = 1.33$	62.54	84.18	28.05	76.95
$\mu = 2.00$	62.86	91.58	52.12	77.20
$\mu = 2.66$	**69.74**	**93.27**	**26.32**	**82.17**
$\mu = 3.00$	65.55	90.57	43.65	79.19
$\mu = 3.99$	66.22	89.23	29.78	79.68
The channel reduction rate of CA				
$r = 1$	63.31	92.59	62.29	77.53
$r = 2$	65.64	91.92	33.78	79.26
$r = 4$	**69.74**	**93.27**	**26.32**	**82.17**
$r = 8$	63.21	90.91	32.64	77.46

VI. CONCLUSION

In this article, we extend the conventional IRSTD task and propose a fuzzy text-guided IRSTD framework to address the issue that current models struggle to further improve detection accuracy by relying only on limited visual information. First, we devise a novel fuzzy text prompt dedicated to IRSTD. Compared to existing explicit prompts, ours can adapt to ambiguous target categories and bridge the text–image semantic gap. Then, the proposed TCMHE can adaptively capture key global and local features through a text-guided CNN-Mamba dual-branch hybrid architecture. Finally, we propose PCSID, which utilizes fuzzy text to guide contextual feature aggregation and multidimensional refinement of feature maps. Moreover, we have constructed the first fuzzy text–image bimodal dataset, called FZDT, to provide the basis for subsequent text-guided IRSTD research. Extensive experimental results demonstrate that our method achieves better detection performance and target contour recovery than the state-of-the-art methods. More surprisingly, the proposed Text-IRSTD demonstrates strong generalization and wide application prospects in unseen detection scenarios. Despite these promising results, some limitations remain, including reliance on manual annotation, limited model efficiency, and frozen text encoders. In future work, we will first explore efficient and accurate automated prompt generation methods to expand application scenarios. Second, model lightweighting techniques will be explored to meet specific high-real-time application requirements. Third, we will explore text encoder fine-tuning or domain adaptation techniques to further improve cross-modal alignment and target detection performance.

REFERENCES

[1] B. Huang, J. Li, J. Chen, G. Wang, J. Zhao, and T. Xu, "Anti-UAV410: A thermal infrared benchmark and customized scheme for tracking drones in the wild," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 2852–2865, May 2024.

[2] F. Lin, K. Bao, Y. Li, D. Zeng, and S. Ge, "Learning contrast-enhanced shape-biased representations for infrared small target detection," *IEEE Trans. Image Process.*, vol. 33, pp. 3047–3058, 2024.

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

[3] Q. Liu, R. Liu, B. Zheng, H. Wang, and F. Ying, "Infrared small target detection with scale and location sensitivity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 17490–17499.

[4] X. Tong et al., "ST-trans: Spatial–temporal transformer for infrared small target detection in sequential images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5001819.

[5] J. Wu et al., "DFINet: Dynamic feedback iterative network for infrared small target detection," *Pattern Recognit.*, vol. 169, Jan. 2026, Art. no. 111958.

[6] M. Zhang, R. Zhang, Y. Yang, H. Bai, J. Zhang, and J. Guo, "ISNet: Shape matters for infrared small target detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 877–886.

[7] Z. Qiu, Y. Ma, F. Fan, J. Huang, and M. Wu, "Adaptive scale patch-based contrast measure for dim and small infrared target detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[8] S. Yuan, H. Qin, X. Yan, N. Akhtar, and A. Mian, "SCTransNet: Spatial-channel cross transformer network for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5002615.

[9] Y.-H. Xin, J. Zhou, and Y.-S. Chen, "Dual multi-scale filter with SSS and GW for infrared small target detection," *Infr. Phys. Technol.*, vol. 81, pp. 97–108, Mar. 2017.

[10] Y. Li, Z. Li, Y. Shen, and J. Li, "Infrared small target detection based on 1-D difference of guided filtering," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[11] Y. Xu et al., "Infrared small target detection based on local contrast-weighted multidirectional derivative," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5000816.

[12] X. Zhang, J. Ru, and C. Wu, "Infrared small target detection based on gradient correlation filtering and contrast measurement," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603012.

[13] L. Chen, T. Wu, S. Zheng, Z. Qiu, and F. Huang, "Robust unsupervised multifeature representation for infrared small target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 10306–10323, 2024.

[14] X. Kong, C. Yang, S. Cao, C. Li, and Z. Peng, "Infrared small target detection via nonconvex tensor fibered rank approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5000321.

[15] Y. Luo, X. Li, S. Chen, C. Xia, and L. Zhao, "IMNN-LWEC: A novel infrared small target detection based on spatial–temporal tensor model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5004022.

[16] F. Yan, G. Xu, J. Wang, Q. Wu, and Z. Wang, "Infrared small target detection via schatten capped pNorm-based non-convex tensor low-rank approximation," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[17] Y. Dai, Y. Wu, F. Zhou, and K. Barnard, "Asymmetric contextual modulation for infrared small target detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 949–958.

[18] Y. Zhang, W. Bao, Y. Yang, W. Wan, Q. Xiao, and X. Zou, "HAFNet: Hierarchical attention fusion network for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5007316.

[19] Q. Hou, Z. Wang, F. Tan, Y. Zhao, H. Zheng, and W. Zhang, "RISTDnet: Robust infrared small target detection network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[20] F. Wu, A. Liu, T. Zhang, L. Zhang, J. Luo, and Z. Peng, "Saliency at the helm: Steering infrared small target detection with learnable kernels," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5000514.

[21] T. Chen et al., "MiM-ISTD: Mamba-in-Mamba for efficient infrared small-target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5007613.

[22] S. Li, Q. Ma, S. Zhang, and C. Yang, "*DCCS-Det: Directional context and cross-scale-aware detector for infrared small target*," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 4400215.

[23] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[24] L. H. Li et al., "Grounded language-image pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10965–10975.

[25] X. Wu, F. Zhu, R. Zhao, and H. Li, "CORA: Adapting CLIP for open-vocabulary detection with region prompting and anchor pre-matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7031–7040.

[26] M. Zhang, Q. Xu, Y. Wang, X. Li, and H. Yuan, "MIRSAM: Multi-modal vision-language segment anything model for infrared small target detection," *Vis. Intell.*, vol. 3, no. 1, pp. 1–13, Dec. 2025.

[27] M. Zhang, X. Li, F. Gao, J. Guo, X. Gao, and J. Zhang, "SAIST: Segment any infrared small target model guided by contrastive language-image pretraining," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 9549–9558.

[28] F. Huang, S. Zheng, Z. Qiu, H. Liu, H. Bai, and L. Chen, "Text-IRSTD: Leveraging semantic text to promote infrared small target detection in complex scenes," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 10635–10644.

[29] H. Zhang et al., "ResNest: Split-attention networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jul. 2022, pp. 2735–2745.

[30] H. Zhu, J. Zhang, G. Xu, and L. Deng, "Balanced ring top-hat transformation for infrared small-target detection with guided filter kernel," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 5, pp. 3892–3903, Oct. 2020.

[31] R. Kou et al., "LW-IRSTNet: Lightweight infrared small target segmentation network and application deployment," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5621313.

[32] H. Sun, J. Bai, F. Yang, and X. Bai, "Receptive-field and direction induced attention network for infrared dim small target detection with a large-scale dataset IRDST," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5000513.

[33] M. Xu, C. Yu, Z. Li, H. Tang, Y. Hu, and L. Nie, "*HDNet: A hybrid domain network with multiscale high-frequency information enhancement for infrared small-target detection*," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5004115.

[34] G. Hu, L. Fan, H. Xu, C. Lin, X. Ding, and Y. Huang, "Dual-transformer feature enhancement for infrared small-dim target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 342–356, 2026.

[35] S. Zhuang, Y. Hou, M. Qi, and D. Wang, "FAA-Net: A frequency-aware attention network for single-frame infrared small target detection," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–16, 2025.

[36] B. Li et al., "Dense nested attention network for infrared small target detection," *IEEE Trans. Image Process.*, vol. 32, pp. 1745–1758, 2023.

[37] T. Yue, X. Lu, J. Cai, Y. Chen, and S. Chu, "SDS-Net: Shallow–deep synergism-detection network for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 3001113.

[38] W. Cai et al., "Frequency-aware lightweight distillation for infrared small target detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 62, pp. 11167–11180, 2026.

[39] X. Wu, D. Hong, and J. Chanussot, "UIU-Net: U-Net in U-Net for infrared small object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 364–376, 2023.

[40] L. Fan et al., "Diffusion-based continuous feature representation for infrared small-dim target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5003617.

[41] Y. Zhang, M. Qi, K. Li, X. Li, and S. Zhuang, "FLINet: Fuzzy logic for infrared small-target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5005216.

[42] M. Zhang, Y. Wang, J. Guo, Y. Li, X. Gao, and J. Zhang, "IRSAM: Advancing segment anything model for infrared small target detection," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 233–249.

[43] Y. Fu, J. Lyu, P. Ma, Z. Liu, and M. K. Ng, "*A unified SAM-guided self-prompt learning framework for infrared small target detection*," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5008014.

[44] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[45] M. Zhang, X. Li, F. Gao, and J. Guo, "IRMamba: Pixel difference Mamba with layer restoration for infrared small target detection," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 9, pp. 10003–10011.

[46] R. Wu, Y. Liu, P. Liang, and Q. Chang, "H-vmunet: High-order vision Mamba UNet for medical image segmentation," *Neurocomputing*, vol. 624, Apr. 2025, Art. no. 129447.

[47] P. Xiao et al., "MF-Mamba: Multiscale convolution and Mamba fusion model for semantic segmentation of remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5405916.

[48] J. Zhang, J. Huang, S. Jin, and S. Lu, "Vision-language models for vision tasks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5625–5644, Aug. 2024.

[49] Y. Zhong et al., "RegionCLIP: Region-based language-image pretraining," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16772–16782.

[50] X. Gu, T.-Y. Lin, W. Kuo, and Y. Cui, "Open-vocabulary object detection via vision and language knowledge distillation," 2021, *arXiv:2104.13921*.

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.

[51] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, "YOLO-world: Real-time open-vocabulary object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16901–16911.
[52] X. Xi, Y. Huang, R. Luo, and Y. Qiu, "OW-OVD: Unified open world and open vocabulary object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 25454–25464.
[53] X. Yi, H. Xu, H. Zhang, L. Tang, and J. Ma, "Text-IF: Leveraging semantic text guidance for degradation-aware and interactive image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27016–27025.
[54] J. Yang, S. Liu, J. Wu, X. Su, N. Hai, and X. Huang, "Pinwheel-shaped convolution and scale-based dynamic loss for infrared small target detection," in *Proc. AAAI Conf. Artif. Intell.*, 2025, pp. 9202–9210.
[55] H. Yang et al., "PBT: Progressive background-aware transformer for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5004513.
[56] B. Yang, X. Zhang, J. Zhang, J. Luo, M. Zhou, and Y. Pi, "EFLNet: Enhancing feature learning network for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5906511.
[57] C. Gao, D. Meng, Y. Yang, Y. Wang, X. Zhou, and A. G. Hauptmann, "Infrared patch-image model for small target detection in a single image," *IEEE Trans. Image Process.*, vol. 22, no. 12, pp. 4996–5009, Dec. 2013.
[58] L. Zhang and Z. Peng, "Infrared small target detection based on partial sum of the tensor nuclear norm," *Remote Sens.*, vol. 11, no. 4, p. 382, Feb. 2019.
[59] Y. Qin, L. Bruzzone, C. Gao, and B. Li, "Infrared small target detection based on facet kernel and random Walker," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7104–7118, Sep. 2019.
[60] Z. Qiu, Y. Ma, F. Fan, J. Huang, and L. Wu, "Global sparsity-weighted local contrast measure for infrared small target detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[61] Y. Dai, Y. Wu, F. Zhou, and K. Barnard, "Attentional local contrast networks for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9813–9824, Nov. 2021.
[62] T. Zhang, L. Li, S. Cao, T. Pu, and Z. Peng, "Attention-guided pyramid context networks for detecting infrared small target under complex background," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 4, pp. 4250–4261, Aug. 2023.
[63] T. Guo, B. Zhou, F. Luo, L. Zhang, and X. Gao, "DMFNet: Dual-encoder multistage feature fusion network for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614214, doi: 10.1109/TGRS.2024.3376382.
[64] Y. Dai et al., "Pick of the bunch: Detecting infrared small targets beyond hit-miss trade-offs via selective rank-aware attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5006215.
[65] Y. Huang, Z. Tang, D. Chen, K. Su, and C. Chen, "Batching soft IoU for training semantic segmentation networks," *IEEE Signal Process. Lett.*, vol. 27, pp. 66–70, 2020.
[66] K. Wu et al., "TinyCLIP: CLIP distillation via affinity mimicking and weight inheritance," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 21913–21923.



Liqiong Chen received the B.S. and Ph.D. degrees from the School of Electronic Information, Wuhan University, Wuhan, China, in 2016 and 2021, respectively.

She is currently an Associate Professor with the School of Mechanical Engineering and Automation, Fuzhou University, Fuzhou, China. Her research interests include object detection, scene recognition, and deep learning.



Yuhao Liu received the B.S. degree from the School of Mechanical and Electronic Engineering, Jingdezhen Ceramic University, Jiangxi, China, in 2024.

He is currently a postgraduate with the School of Mechanical Engineering and Automation, Fuzhou University, Fuzhou, China. His research interests focus on deep learning and infrared small target detection.



Zhaobing Qiu received the B.S., M.S., and Ph.D. degrees from the School of Electronic Information, Wuhan University, Wuhan, China, in 2016, 2019, and 2023, respectively.

He is currently an Associate Research Fellow with the School of Mechanical Engineering and Automation, Fuzhou University, Fuzhou, China. His research interests include computer vision, pattern recognition, and optomechanical system control.



Shuyuan Zheng received the B.S. degree from the School of Electromechanical and Automation, Huaqiao University, Xiamen, China, in 2023.

She is currently a postgraduate with the School of Mechanical Engineering and Automation, Fuzhou University, Fuzhou, China. Her research interests focus on deep learning and infrared small target detection.



Jiaxin Zou received the B.S. degree from the School of Marine Equipment and Mechanical Engineering, Jimei University, Xiamen, China, in 2025.

He is currently a postgraduate with the School of Mechanical Engineering and Automation, Fuzhou University, Fuzhou, China. His research interests focus on deep learning and infrared small target detection.



Yangping Yuan received the B.S. degree from Southwest University, Chongqing, China, in 2015, and the M.S. degree from Xiamen University, Xiamen, China, in 2018.

He is currently an Algorithm Engineer with Zhipu AI, Beijing, China. His research interests include natural language processing, autonomous agents, computer vision, and large language models.



Feng Huang was born in Fuzhou, China, in 1979. He is currently pursuing with the School of Mechanical Engineering and Automation, Fuzhou University, Fuzhou.

His research interests focus on optoelectronic imaging.

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on October 17,2026 at 07:42:58 UTC from IEEE Xplore. Restrictions apply.